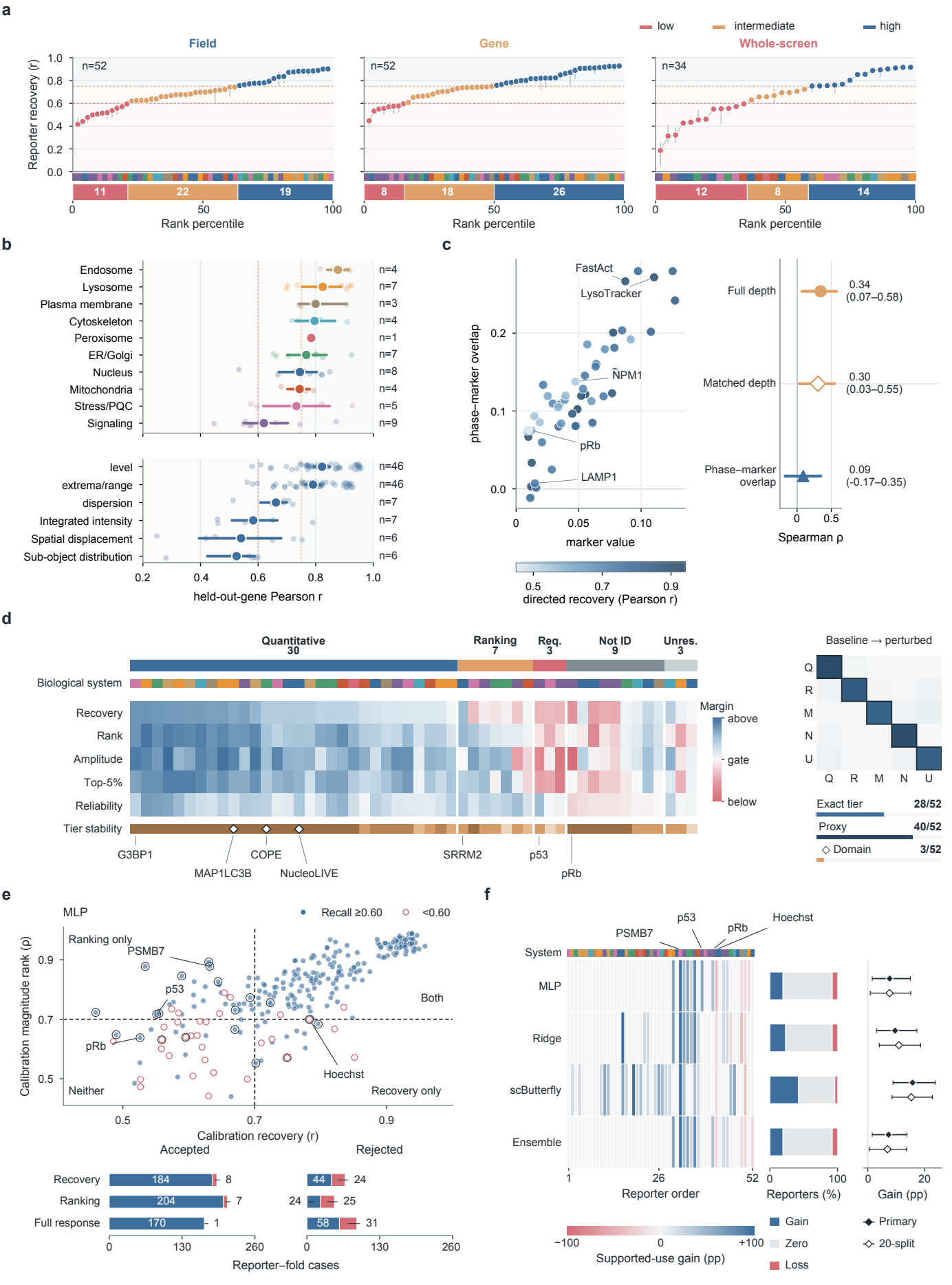

a
low
intermediate
high
Field
Gene
Whole-screen
Reporter recovery (r)
n=52
n=52
n=34
11
22
19
8
18
26
12
8
14
Rank percentile
b
Endosome
Lysosome
Plasma membrane
Cytoskeleton
Peroxisome
ER/Golgi
Nucleus
Mitochondria
Stress/PQC
Signaling
level
extrema/range
dispersion
Integrated intensity
Spatial displacement
Sub-object distribution
held-out-gene Pearson r
c
FastAct
LysoTracker
NPM1
pRb
LAMP1
phase–marker overlap
marker value
directed recovery (Pearson r)
Full depth
Matched depth
Phase–marker overlap
0.34 (0.07–0.58)
0.30 (0.03–0.55)
0.09 (-0.17–0.35)
Spearman ρ
d
Quantitative 30
Ranking 7
Req. 3
Not ID 9
Unres. 3
Biological system
Recovery
Rank
Amplitude
Top-5%
Reliability
Tier stability
Margin
above
gate
below
G3BP1
MAP1LC3B
COPE
NucleoLIVE
SRRM2
p53
pRb
Baseline → perturbed
Q
R
M
N
U
Exact tier 28/52
Proxy 40/52
Domain 3/52
e
MLP
Recall ≥0.60
<0.60
Ranking only
PSMB7
p53
pRb
Both
Hoechst
Neither
Recovery only
Calibration magnitude rank (ρ)
Calibration recovery (r)
Accepted
Rejected
Recovery
Ranking
Full response
184
8
204
7
170
1
44
24
24
25
58
31
Reporter–fold cases
f
PSMB7
p53
pRb
Hoechst
System
MLP
Ridge
scButterfly
Ensemble
Reporter order
Reporters (%)
Gain (pp)
Supported-use gain (pp)
Gain
Zero
Loss
Primary
20-split